# Prospective Remeasurement Alerts for Suspicious Pediatric Growth Measurements in Electronic Health Records

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## Abstract

**Objective.** Retrospective identification of measurement errors in pediatric EHR growth data is limited because the original clinical measurement usually cannot be repeated after a value is saved. We frame the task prospectively: rather than adjudicating historical records alone, a real-time alert fires at the moment of height entry — while the child is still present and remeasurement remains possible — turning uncertain retrospective judgments into opportunities for immediate repeat measurement. We developed and evaluated a multi-detector, confidence-fused remeasurement-alert pipeline for pediatric height data in simulated streaming evaluation and retrospective EHR alert-burden analysis.

**Materials and Methods.** The pipeline combines two hard rules (biologically implausible value thresholds; carry-forward duplicate detection) with five soft detectors (short- and long-interval velocity z-scores, height-for-age z-score change, streaming LOWESS trajectory residual, soft HAZ magnitude), fused via weighted noisy-OR into a single score; an alert fires when this score exceeds a tunable threshold (default 0.60). We evaluated the pipeline in streaming mode — using only prior unflagged history per visit, mirroring live deployment — on six published Monte Carlo simulated-error benchmarks (Harrall et al.; 600 patients, 17,705 visits), comparing it against streaming reimplementations of the Daymont and Harrall algorithms and the EPIC 10% rule. Real-world alert burden was characterized in a de-identified outpatient pediatric EHR extract from the Pediatric Physicians' Organization at Children's (PPOC; 250,588 patients, ∼6.5M visits).

**Results.** At the default threshold, the pipeline achieved pooled sensitivity 0.88 and specificity 0.96 — exceeding streaming Harrall (0.71 / 0.95) and Daymont (0.82 / 0.99) on sensitivity, with a modest specificity tradeoff versus Daymont that is continuously adjustable via threshold. The EPIC 10% rule's sensitivity of 0.76 came at a specificity of 0.33, producing ∼1.76M alerts (∼27% of PPOC visits); at the default threshold, the proposed pipeline matches that sensitivity floor while achieving 0.96 specificity, and at threshold 0.70 reduces alert burden to ∼18% of visits with sensitivity 0.86.

**Conclusion.** In simulated streaming evaluation, a prospective, trajectory-aware remeasurement-alert pipeline achieved higher sensitivity than streaming comparator implementations, lower specificity than Daymont, and much higher specificity than the EPIC 10% rule. Real-world EHR analysis

suggested substantially lower alert burden than the EPIC rule, with threshold-dependent tradeoffs. Prospective clinical validation, in which alerts trigger immediate remeasurement while the child is present, is needed to determine real-world predictive value, workflow impact, and clinical utility.

# 1 Background

A central obstacle in pediatric growth-data quality research is that the true value of a clinical measurement is usually unrecoverable once it has been saved to the electronic health record. A recorded height of 112 cm for a seven-year-old may reflect a genuine measurement, a transcription error, a unit confusion, or a carry-forward duplication of a prior visit's value — and these possibilities cannot be reliably distinguished from the longitudinal record alone, even by trained clinicians with full access to the child's history. This is not a contingent limitation of existing methods; it is a structural property of retrospective EHR data. As Daymont noted, the same record contains both plausible outliers (true extreme values that look like errors) and implausible inliers (erroneous values that look ordinary against the population distribution but are wrong for the individual child) [7], and retrospective procedures have limited ability to separate these categories without an external reference measurement.

A substantial and sophisticated literature has developed retrospective cleaning algorithms that, given a child's full longitudinal record, attempt to declare each historical measurement plausible or implausible [1,2,6,9,10,11]. These algorithms are evaluated on either Monte Carlo simulated data with known injected errors [9] or human consensus labels whose inter-rater agreement is substantially below 1.0 [1] — not because the field has been methodologically incautious, but because no better evidence base exists. The most widely adopted algorithm, Daymont's growthcleanr [1], has been validated against physician judgment and applied to large multisite consortia [4]; more recent approaches by Shi et al. [10], Phan et al. [6], and Harrall et al. [9] have introduced robust regression, peak-removal, and semi-parametric trajectory smoothing. All of these advances operate within the same epistemological constraint.

Real-time clinical decision support in this domain is comparatively underdeveloped. The widely deployed EPIC EHR system alerts clinicians when a newly entered height differs from the immediately preceding visit by more than 10%, and the clinician can either save or correct the value. As we show below, this rule has poor specificity — it flags an enormous fraction of plausible growth — and its anchoring to a single prior point makes it vulnerable to error propagation when the prior point was itself wrong. To our knowledge no commercial EHR currently deploys a trajectory-aware alert.

This paper proposes a resolution to the identifiability problem that does not require retrospective labels. The key insight is that the problem is not intrinsic to the task of detecting measurement errors — it is intrinsic to the *timing* of the detection attempt. At the moment a new height is entered, while the child is still in the clinic, the ground truth remains recoverable: a suspicious measurement can be remeasured, and the clinician and patient are present to resolve the ambiguity. We therefore reframe the task not as "is this historical measurement an error?" — a question that has no recoverable answer — but as "is this new measurement suspicious enough that we should remeasure it right now?" — a question that can be answered by action. This shift does not prove error status retrospectively, but it creates an opportunity to obtain a reference measurement while the ambiguity can still be acted on. The deployment failure modes are also asymmetric in a clinically important way: a missed alert leaves the EHR in the same state it would otherwise be in, while a false alert costs a thirty-second remeasurement.

# 2 Methods

## 2.1 Conceptual framing

The pipeline runs at the moment a new height is entered, while the child is still in the clinic. Every detector sees only the new measurement and the child's prior history — no future points, in contrast to retrospective cleaning algorithms [1,9].

For each entered measurement the pipeline outputs a continuous error confidence in [0, 1] and a categorical label: error (population BIV or exact carry-forward duplicate), remeasure (fused confidence $\geq$ alert threshold), or no alert. The clinician decides what to do with an alert — remeasure or keep — and the saved value is their choice.

For evaluation, we adopt one simplifying assumption: every alerted point is removed from the history available to subsequent points, equivalent to assuming each alert is acted on and resolved (either remeasured to truth or discarded). The same convention is applied to every comparator.

## 2.2 Pipeline overview

The pipeline applies two hard rules and five soft detectors in sequence. The hard rules immediately raise an "error" alert if triggered. Otherwise, the five soft detectors are evaluated, each producing a confidence in [0, 1], and the five confidences are fused with a weighted noisy-OR rule [14] into a single error confidence. A "remeasure" alert is issued when the fused confidence clears the configured threshold.

## 2.3 Hard rules

**Population BIV check.** Each new height is converted to a sex- and age-specific z-score using WHO Child Growth Standards for ages $\leq$ 1856 days [12] and CDC growth charts for ages $>$ 1856 days [13]. For ages in the WHO range an absolute HAZ $\geq$ 6 raises an error. In the CDC range the modified HAZ recommended by the CDC for extreme-value detection is used, with the CDC-recommended cutoffs of mod_HAZ $<$ -5 or mod_HAZ $>$ 4 [13].

**Carry-forward duplicate check.** A new height that matches the immediately previous observed height to within $1\times10^{-6}$ cm raises an error. Exact equality between consecutive heights is essentially never the result of two independent measurements and is one of the most common error modes in EHR growth data — Lin et al. report that carry-forward measurements were the most common error mode flagged in the PCORnet pediatric cohort, accounting for the majority of flagged pediatric heights [4]. Note that this check compares against the most recent *observed* prior point, not the most recent *un-flagged* point, so that a prior error followed by a carry-forward replication of that prior error is still caught.

## 2.4 Soft detectors

If the hard rules do not fire, the pipeline evaluates the following five detectors. All of them depend only on the new point and the patient's prior un-flagged history.

**Velocity (adjacent).** The short-interval velocity is computed as the change in height between the new point and the immediately preceding visit, divided by the age difference. Velocity is converted to an age- and sex-specific z-score against velocity reference distributions derived by numerical differentiation of the WHO and CDC LMS tables [12,13]. Adjacent velocity is the detector most sensitive to single-point errors: an isolated erroneous spike inflates the velocity both going in and coming out, so the adjacent-velocity z-score after the error is large in magnitude. Its threshold is $|z| > 5$.

**Velocity (adjusted).** The same velocity z-score, but computed against the most recent prior visit whose interval meets an age-specific minimum (90 days for ages 0–12 months, 180 days for ages 1–2 years, 335 days for ages 2–12 years, 180 days for ages 13–18 years, reflecting the shortest interval over which measurement noise is small relative to true growth). The adjusted velocity catches sustained shifts but, importantly, *dilutes* a single-point error toward the population mean when the interval used is long.

**HAZ change.** The absolute change in HAZ between the new point and the immediately previous visit is compared to a dynamic, age- and interval-dependent plausibility threshold. This threshold is wider for younger children and longer intervals (during which legitimate channel-crossing growth shifts are larger) and narrower for older children and shorter intervals.

**Streaming LOWESS trajectory residual.** A robust LOWESS curve [15,16] is fit to the HAZ values in the patient's prior history (minimum five points) and extrapolated forward to the age of the new point. The residual — new HAZ minus predicted HAZ — is converted to a robust z by dividing by a residual scale, and the detector fires when $|z| > 3$.

The residual scale is estimated from self-free k-nearest-neighbor-median residuals: each history point is compared to the median HAZ of its five nearest-in-age neighbors (excluding itself), and the scale is $1.4826 \times$ MAD of those residuals [17]. The scale is then inflated by a $\sqrt{1 + \mathrm{gap}/\tau}$ factor with $\tau = 730$ days when the new point lies well beyond the last history point, reflecting the fact that HAZ trajectories can legitimately drift over multi-year gaps even in healthy children.

**Soft HAZ magnitude.** A check on the magnitude of the new HAZ alone, threshold $|\mathrm{HAZ}| > 3$, with a deliberately weak weight and a slow confidence ramp. We include this as a "weak prior" — large HAZ values are more often than not real (Freedman et al. [8] cautioned exactly against treating extreme HAZ as automatic evidence of error) but are weakly informative when several other detectors also fire.

## 2.5 Per-detector confidence mapping

A detector that has not crossed its threshold contributes no evidence. Above the threshold, the per-detector confidence ramps smoothly toward 1 as a sigmoid of the *exceedance ratio* $r = |\mathrm{metric}| / \mathrm{threshold}$:

$$c(r) = \begin{cases} 0, & r \leq 1 \\ \dfrac{1}{1 + \exp[-k(r - r_{\text{mid}})]}, & r > 1 \end{cases}$$

with k = 3.0 and a detector-specific $r_{\text{mid}}$. The exceedance form (rather than a sigmoid of the raw deviation) makes the curve scale-invariant across detectors with very different units (velocity z, HAZ, HAZ-change, trajectory z). The $r_{\text{mid}}$ parameter controls how far past the threshold a detector must reach before it contributes meaningful evidence: $r_{\text{mid}} = 1$ would make a just-crossed threshold already score 0.5; we use $r_{\text{mid}} = 1.8$ for the velocity and HAZ-change detectors and $r_{\text{mid}} = 2.0$ for the soft-HAZ and trajectory detectors.

### 2.6 Noisy-OR fusion

Per-detector confidences are pooled by weighted noisy-OR [14], the standard probabilistic combination rule for independent evidence:

$$C_{ensemble} = 1 - \prod_i (1 - w_i \cdot c_i)$$

with detector weights chosen to reflect each detector's expected reliability (adjacent velocity 0.85, adjusted velocity 0.90, HAZ change 0.90, trajectory 0.85, soft HAZ 0.45). This generalizes a "k-of-N voting" rule smoothly: any single detector with a high confidence can drive $C_{\text{ensemble}}$ high, multiple weak detectors can also accumulate to a high $C_{\text{ensemble}}$, and a detector that did not cross its threshold contributes nothing.

A "remeasure" alert is raised when $C_{\text{ensemble}} \geq \tau$, where $\tau$ is a tunable threshold (default 0.60). The threshold sweep in Section 3.2 characterizes its tradeoff.

### 2.7 Streaming application to retrospective benchmarks

To evaluate the alert pipeline on retrospective benchmark data, we apply it in *streaming* mode. For each patient, measurements are sorted in chronological order. The algorithm walks forward through the visits; for each visit, the history available to the detectors is the set of all prior un-flagged measurements for that patient. Once a measurement is flagged (as either "error" or "remeasure"), it is removed from the history that subsequent points are judged against.

This streaming protocol is the faithful simulation of the live-EHR deployment: at the moment of entry, the algorithm has the patient's curated prior history and is asked whether the new point should trigger an alert. We use the same protocol below to evaluate the Daymont and Harrall algorithms for fair comparison.

### 2.8 Evaluation data

We use the Monte Carlo simulated-error datasets released with the Harrall paper [9], which were generated by:

1. Simulating clean longitudinal height trajectories for 100 children each, from a Preece–Baines [18] parametric growth model with three parameter sets corresponding to "low," "medium," and "high" populations (positioned at the 5th–25th, 25th–75th, and 75th–95th CDC percentile bands, respectively).

2. Generating 100 visits per child with mean and variance of visit spacing matched to the EPOCH cohort [19].

3. Injecting three classes of errors at known positions and magnitudes:

- **Cups** — abrupt downward height excursions that recover at the next visit (e.g., a transcription mistake);
- **Caps** — abrupt upward height excursions that recover at the next visit;
- **Repeats** — exact duplicates of the previous visit's height value, simulating EHR carry-forward auto-population.

4. Varying the error rate across two levels labeled "dirt2" (matching the empirical error proportions estimated from the EPOCH cohort) and "dirt3" (double the empirical error rate).

The released benchmark thus consists of six labeled datasets — pop{1,2,3}_dirt{2,3} — with bad_ht = True marking every injected error. Table 1 summarizes these benchmarks; total visits range from 2,818 to 3,022 per dataset, and the simulated error prevalence ranges from 10.4% to 16.2%.

**Table 1.** Composition of the six Harrall simulated-error benchmarks.

| Dataset | Patients | Visits | Injected errors | Error prevalence |
|---|---:|---:|---:|---:|
| pop1_dirt2 | 100 | 2,932 | 362 | 12.3% |
| pop1_dirt3 | 100 | 2,987 | 475 | 15.9% |
| pop2_dirt2 | 100 | 2,960 | 308 | 10.4% |
| pop2_dirt3 | 100 | 2,818 | 457 | 16.2% |
| pop3_dirt2 | 100 | 2,986 | 324 | 10.9% |
| pop3_dirt3 | 100 | 3,022 | 452 | 15.0% |
| **Total** | **600** | **17,705** | **2,378** | **13.4%** |

### 2.9 Comparators

We compare to three streaming baselines:

- **Streaming Daymont.** The Daymont weighted-moving-average algorithm [1] (growthcleanr) applied to each successive truncation of a patient's history, so that the decision for visit $t$ uses only history up to visit $t$ - 1.
- **Streaming Harrall.** The Harrall trajectory-cleaning algorithm [9] applied to each successive history truncation.

- **EPIC 10% rule.** A new measurement is flagged when its height differs from the most recent un-flagged previous height by more than 10%, mimicking the rule deployed in the EPIC EHR. Flagged points are removed from history for subsequent checks.

The Daymont and Harrall algorithms were not designed for prospective use; we include them for completeness so that we can quantify how much each loses by removing bidirectional context.

# 3 Results

## 3.1 Performance at the default threshold

Table 2 reports per-dataset sensitivity and specificity for all four algorithms at the default alert threshold (0.60 for our pipeline, 0.10 for EPIC).

**Table 2.** Sensitivity / specificity by dataset and method, streaming mode.

| Dataset | Harrall (streaming) | Daymont (streaming) | EPIC 10% | This work |
|---|---|---|---|---|
| pop1_dirt2 | 0.68 / 0.94 | 0.79 / 0.99 | 0.73 / 0.39 | **0.87 / 0.97** |
| pop1_dirt3 | 0.75 / 0.96 | 0.85 / 1.00 | 0.77 / 0.28 | **0.91 / 0.96** |
| pop2_dirt2 | 0.67 / 0.96 | 0.80 / 0.99 | 0.77 / 0.30 | **0.87 / 0.94** |
| pop2_dirt3 | 0.70 / 0.91 | 0.82 / 0.98 | 0.81 / 0.28 | **0.86 / 0.93** |
| pop3_dirt2 | 0.72 / 0.97 | 0.80 / 0.99 | 0.73 / 0.36 | **0.87 / 0.98** |
| pop3_dirt3 | 0.72 / 0.95 | 0.83 / 0.99 | 0.74 / 0.36 | **0.89 / 0.96** |
| **Pooled** | **0.71 / 0.95** | **0.82 / 0.99** | **0.76 / 0.33** | **0.88 / 0.96** |

The EPIC 10% rule has by far the lowest specificity (0.33 pooled), reflecting the well-known over-flagging of normal pediatric growth by a fixed 10% rule — adolescent growth spurts and early-childhood velocity routinely produce $> 10\%$ height changes per visit. The Harrall algorithm loses substantial sensitivity in streaming mode (0.71 pooled, compared with $> 0.99$ reported in its retrospective evaluation [9]); the design of its peak-removal step relies on the existence of a downward "cup" *after* a flagged "cap," and that future point is unavailable in streaming use. The Daymont algorithm's weighted moving average uses inverse-distance weighting that gives current and recent points more weight, which translates better to streaming use and degrades less (pooled sensitivity 0.82, specificity 0.99). Daymont also retains the highest specificity of all four methods, exceeding our pipeline's 0.96 by 3 percentage points. Our pipeline achieves the highest pooled sensitivity (0.88) of the four, with specificity (0.96) competitive with the retrospective-style baselines and far above EPIC.

## 3.2 Threshold sensitivity sweep

To characterize the alert threshold's tradeoff, we re-ran the streaming pipeline on all six datasets at remeasure-confidence thresholds 0.30 through 0.90 in increments of 0.10. Because a flagged point is removed from history for subsequent decisions, each threshold is a full re-run, not a post-hoc re-thresholding of cached scores.

**Table 3.** Pooled sensitivity / specificity across all six datasets vs. alert threshold.

| Threshold | Alerted | TP | FP | Sensitivity | Specificity |
|---:|---:|---:|---:|---:|---:|
| 0.30 | 3,727 | 2,115 | 1,612 | 0.89 | 0.89 |
| 0.40 | 3,261 | 2,106 | 1,155 | 0.89 | 0.92 |
| 0.50 | 3,105 | 2,098 | 917 | 0.88 | 0.94 |
| **0.60** | **2,759** | **2,090** | **669** | **0.88** | **0.96** |
| 0.70 | 2,539 | 2,078 | 461 | 0.88 | 0.97 |
| 0.80 | 2,410 | 2,060 | 350 | 0.87 | 0.98 |
| 0.90 | 2,131 | 1,894 | 237 | 0.80 | 0.98 |

The tradeoff is asymmetric. Between thresholds 0.30 and 0.80, sensitivity declines only modestly (0.89 → 0.87); the false-positive count falls almost five-fold (1,611 → 347) for the cost of 54 true positives (2.5% relative loss in TP). Only beyond threshold 0.80 does sensitivity drop steeply. The implication for deployment is that a clinic with low tolerance for false alarms can move the threshold up to 0.80 with little loss of error capture, and only at the most conservative end ($\geq 0.90$) does the algorithm begin to miss meaningfully more true errors.

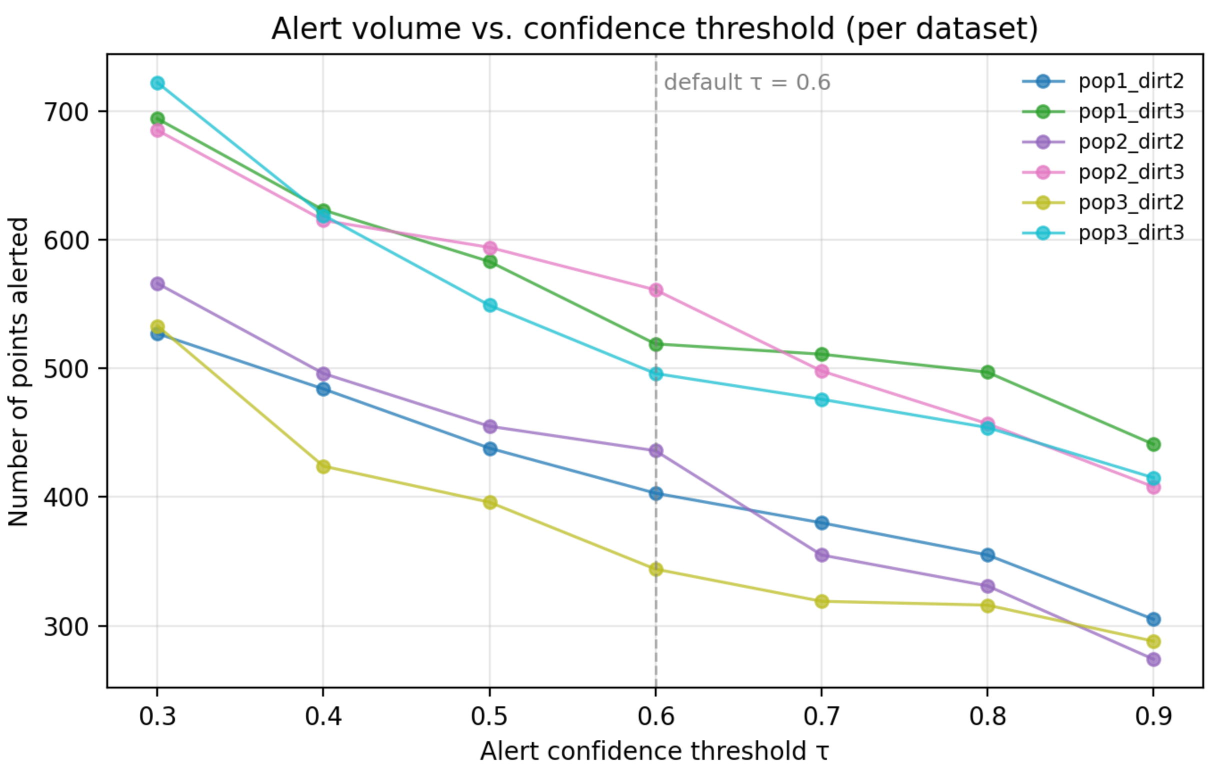


Figure 1: Number of points alerted as a function of the confidence threshold $\tau$, shown separately for each of the six simulated-error datasets. The dashed vertical line marks the default threshold ($\tau = 0.60$).

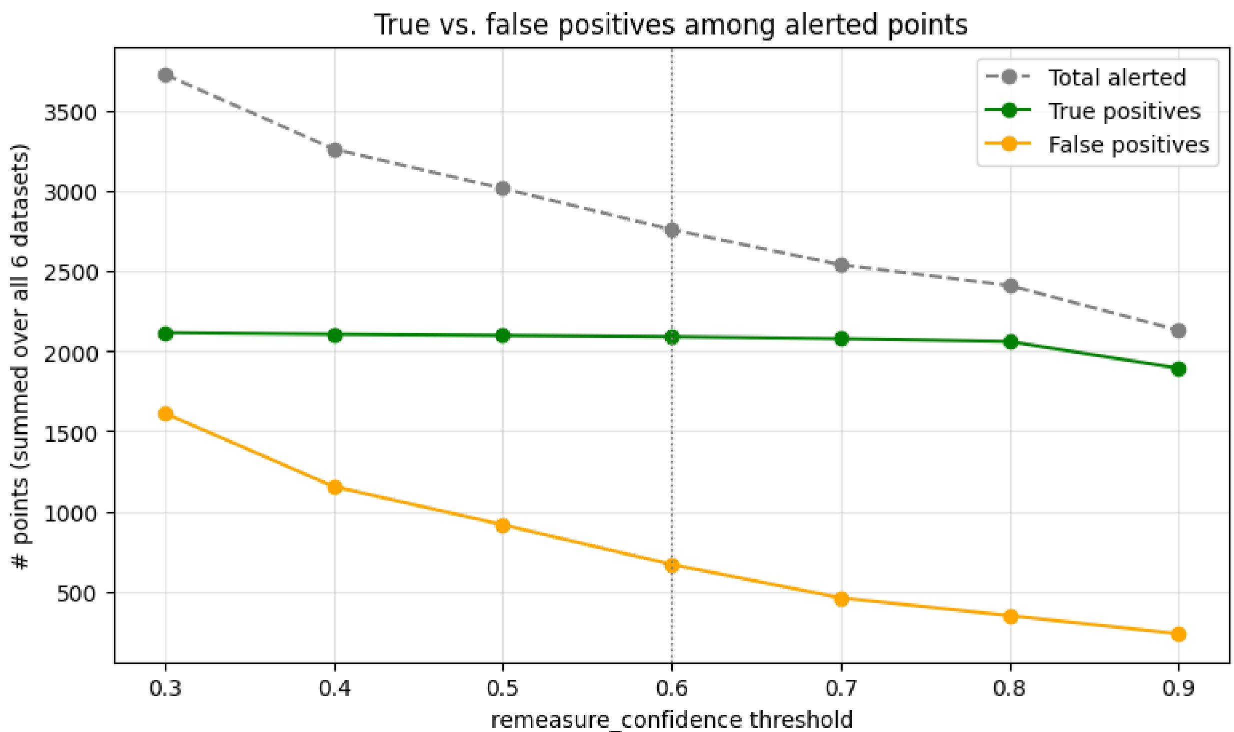


Figure 2: Alerted points pooled across all six datasets, decomposed as a function of the confidence threshold $\tau$. The dashed gray line is the total number of points alerted; the green line is the subset that are true positives (correctly flagged simulated errors); the orange line is the subset that are false positives (clean points incorrectly flagged).

## 3.3 Operational alert burden on a real-world EHR cohort

**Dataset.** We used a de-identified longitudinal EHR extract from the Pediatric Physicians' Organization at Children's (PPOC), a primary-care network affiliated with Boston Children's Hospital. The cohort comprises 250,588 unique pediatric patients aged 0–18 years, with 6,494,473 clinical visits ($\approx$ 72.8% routine office visits) recorded across the network. Each visit row carries patient demographics (sex, ethnicity, race), encounter type, anthropometrics (height, weight, BMI and derived percentiles), and up to 33 ICD-10 encounter diagnoses; the broader extract also includes 17.2 M laboratory result components, 3.8 M medication records, 1.7 M problem-list entries, and 349,827 specialty referrals, all linked by `patient_id` and `visit_id`. Dates are de-identified as `age_in_days`. For the present analysis we used only the height column from the visits table. Unlike the Harrall simulator, no ground-truth error labels are available, so this analysis reports alert volume — the count and share of measurements that would trigger an alert — rather than sensitivity or specificity.

**Analysis.** To characterize the operational burden the two approaches would impose in live clinical use, we ran both the EPIC 10% rule and our pipeline (in streaming mode, across the same threshold sweep used in Section 3.2) on every height measurement in the cohort, removing flagged points from each patient's running history as in the simulated-benchmark protocol.

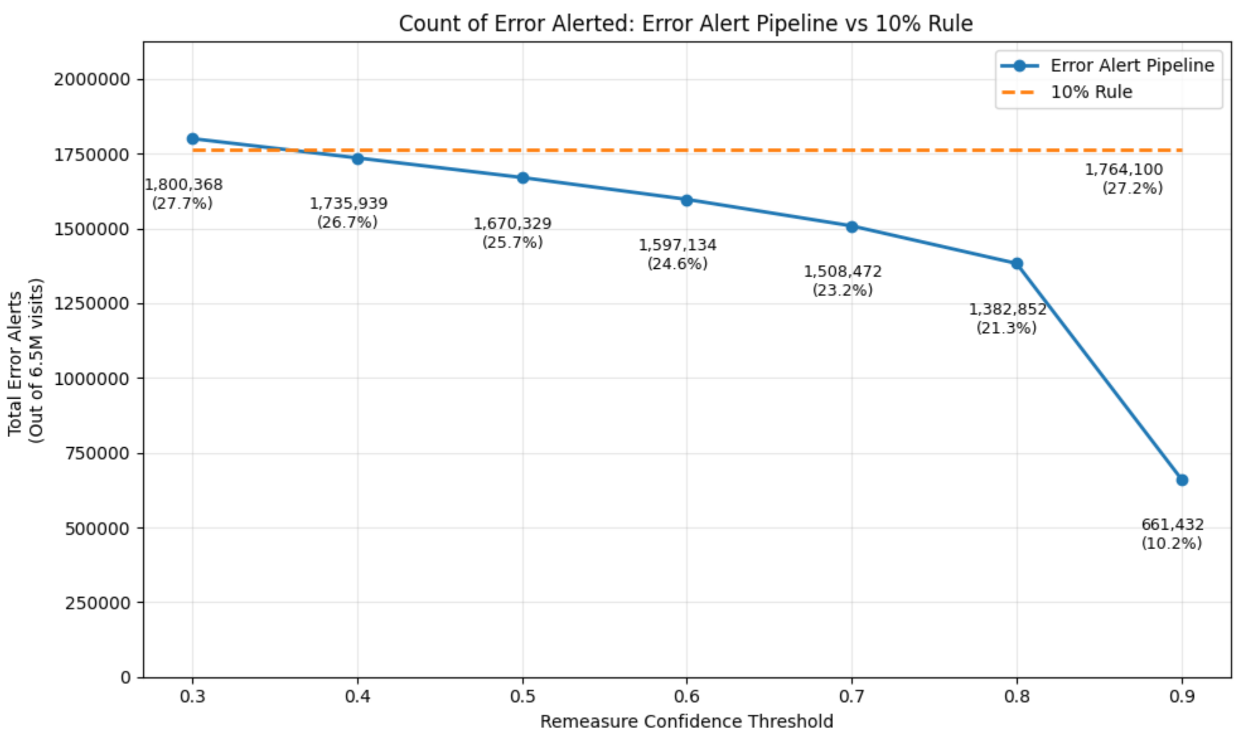


Figure 3: Total number of height measurements that would trigger an alert in the PPOC cohort ($\approx$ 6.5 M visits) under the EPIC 10% rule (orange dashed) and under the error alert pipeline at remeasure-confidence thresholds from 0.30 to 0.90 (solid blue). Annotations report absolute counts and the percentage of the full cohort.

The EPIC 10% rule alerts on roughly 1.76 million measurements ($\approx$ 27% of all visits). Across the evaluated thresholds, the pipeline would have generated far fewer alerts in this retrospective EHR extract, and the threshold sweep quantifies the tradeoff between simulated-error sensitivity and expected remeasurement burden. These results suggest that trajectory-aware alerting could reduce alert volume relative to the EPIC rule, but prospective deployment is needed to measure adherence, PPV, false negatives, and workflow impact in practice.

# 4 Discussion

## 4.1 Why a prospective frame

The main contribution of this paper is a prospective problem formulation for pediatric growth-data quality. Retrospective growth-data cleaning methods operate under an important evidentiary constraint: once a measurement has been saved, the original physical measurement cannot usually be repeated under the same clinical circumstances. Simulated benchmarks and human consensus labels are therefore useful but incomplete evaluation tools for methods intended to identify real clinical measurement errors.

Prospective alerting changes the available evidence. If an alert fires while the child is still present, immediate remeasurement can provide a contemporaneous reference measurement and can directly test whether the alert identified a value that would change with repeat measurement. This does not eliminate all uncertainty, because remeasurement itself has error and workflow adherence may vary, but it provides a practical path to stronger external validation than retrospective record review alone.

The simulated benchmark used in this paper is the same evidentiary foundation on which the retrospective cleaning literature rests. Our evaluation should therefore be interpreted as simulated-benchmark evidence: useful for comparing methods under controlled error assumptions, but insufficient to establish clinical superiority in real-world deployment. The performance results in Section 3 should be read as establishing that the algorithm meets a benchmark used by prior algorithms in this space — a necessary condition for serious consideration — rather than as a definitive demonstration of clinical superiority. Section 4.4 outlines the experiment that would establish the latter.

## 4.2 Comparison to EPIC 10% rule

The EPIC 10% rule's low specificity in the Harrall simulated-error benchmark (0.33 pooled) reflects a mismatch between a fixed proportional threshold and pediatric growth physiology. Children grow at strongly age-dependent velocities, and 10% per visit is easily exceeded by ordinary growth in infants ($\approx$ 50% per year), early adolescents during peak height velocity, and any child with intervisit gaps of more than a year. Our results suggest that even a small amount of trajectory awareness — at minimum, age- and sex-adjusted velocity z-scores — could substantially reduce alert burden in EHRs that currently deploy the 10% rule, without sacrificing the sensitivity to gross errors that motivated the rule in the first place.

## 4.3 Limitations

The primary limitation of this evaluation — that it relies on simulated data — reflects a structural challenge for pediatric EHR measurement-error research. Once a measurement has been saved, the original clinical measurement usually cannot be repeated under the same circumstances, so retrospective studies must rely on simulated errors, consensus labels, or other imperfect reference standards. This paper uses that same evidentiary base. The prospective framing is valuable because it could collect contemporaneous repeat measurements when an alert fires, allowing future studies to estimate predictive value, adherence, and workflow impact directly in clinical use; sensitivity would require additional repeat measurement of a sample of unalerted visits, as outlined below.

**No weight evaluation.** This work covers only height; weight is more difficult because legitimate short-term decreases are common and would require different velocity and HAZ-change references. Extension to weight is a natural next step.

**Static thresholds and weights.** The detector weights and $r_{\text{mid}}$ parameters are hand-chosen based on simulator-tuning and clinical reasoning; they could in principle be learned from a labeled training set. We did not pursue this because the Harrall simulator's error distribution is itself synthetic and learning weights on it risks overfitting to assumptions that do not hold in real data.

## 4.4 Future directions

The arguments in Section 4.1 and Section 4.3 motivate a prospective in-clinic study in which the algorithm fires, the child is remeasured before they leave the room, and the original–remeasurement pair is recorded for evaluation. This design would estimate PPV, sensitivity, adherence, and workflow burden using contemporaneous repeat measurements rather than retrospective labels alone. We outline below a protocol intended to support a multi-site evaluation.

**Study design.** A prospective, multi-site observational deployment study in pediatric primary-care clinics with EHR-integrated growth charting (e.g., participating PPOC sites). The pipeline runs as a real-time decision support widget invoked at the moment a height is committed in the EHR. When the fused confidence exceeds the deploying clinic's threshold (default 0.60), the medical assistant or nurse who recorded the initial measurement is prompted, before the visit closes, to repeat the measurement using a standardized protocol. Both values, the time elapsed between them, the firing detectors, and the fused confidence are written to a study log; the EHR-saved value remains the clinician's choice and is also logged. Independent of alert firing, a random 1–2% sample of unalerted measurements is also remeasured to estimate the false-negative rate.

**Primary endpoints.** (i) Positive predictive value (PPV) of the alert at the default threshold, with 95% confidence interval; (ii) PPV as a function of the confidence threshold across the same sweep used in Section 3.2; (iii) sensitivity, estimated from the random-sample remeasurement arm.

**Clinician workflow and adherence.** The alert appears as a non-blocking modal immediately on save, with a one-line rationale derived from the firing detectors (e.g., "implausible velocity since previous visit; HAZ change above age-adjusted threshold"). The remeasurement step is logged whether or not it is performed; non-adherence is itself a primary descriptive outcome, since a CDS alert that is consistently dismissed is operationally a failed alert regardless of its statistical performance.

This protocol is, in our view, the appropriate next study. The simulated benchmarks in Section 3 establish that the algorithm is competitive in the regime where ground truth is artificially available; the PPOC analysis in Section 3.3 establishes that it produces a tractable alert burden in real EHR traffic; what remains is to establish, on real patients, that the alerts it does produce correspond to measurements a careful remeasurement would in fact change.

Two extensions are worth flagging beyond the prospective study above. First, extending the same pipeline to weight, with appropriate reference standards and an additional set of detectors that handle plausible short-term decreases. Second, exploring representation choices for downstream learning: whether large language models or other learned models perform better when growth data are presented as raw measurements, z-scores, percentiles, velocity features, or growth-chart images.

# 5 Conclusion

The central contribution of this paper is a prospective framing of the growth-data quality problem. Retrospective adjudication of EHR measurement quality is constrained because the original clinical measurement usually cannot be repeated after the fact, leaving simulated errors and human consensus labels as imperfect but useful references. A prospective alert, by contrast, fires at the moment of entry, while the child and clinician are present and remeasurement is possible. This framing turns some uncertain retrospective judgments into actionable opportunities for repeat measurement and creates a path for stronger clinical validation.

Within this framing, the multi-detector confidence-fused pipeline presented here achieved pooled sensitivity 0.88 and specificity 0.96 on the Harrall Monte Carlo benchmark in streaming mode, with higher sensitivity than streaming Harrall and Daymont implementations, lower specificity than Daymont, and much higher specificity than the EPIC 10% rule. In the PPOC EHR extract, the same thresholds would have produced far fewer alerts than the EPIC rule. These findings support prospective evaluation of trajectory-aware remeasurement alerts, but do not by themselves establish clinical effectiveness, workflow acceptability, or generalizability across EHR implementations.

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